

# Hackathon

Development of low cost automation solutions for Construction Tasks.

## **Jointly hosted by**

Ratan Tata Innovation Hub (RTIH)

## Background & Context

The construction industry is a cornerstone of infrastructure development, yet it continues to face challenges of labor-intensive processes, inconsistent quality, and safety risks. Tasks such as rebar tying, welding, and concrete pouring are often prone to delays, human error, and hazardous working conditions.

Globally, advanced robotics and automation technologies are transforming construction by enabling faster execution, precision-driven quality, and enhanced worker safety. In India, the adoption of such technologies can accelerate infrastructure projects, reduce costs, and improve compliance with safety standards.

Given this backdrop, there is a strong need for scalable, modular, and secure robotics solutions that integrate seamlessly into construction workflows, ensuring efficiency, consistency, and safety across diverse project sites.

## Challenge Overview

The Hackathon invites startups, innovators, institutions, and technology providers to design and develop advanced robotics and automation solutions capable of performing otherwise laborious construction tasks

The proposed solution should be suitable for deployment across public infrastructure projects, private construction firms, and government-led housing and development programs.

# I. Challenge Vision and Rationale

## Overarching Goal

To develop robotics-driven construction solutions that deliver faster execution, consistent quality, and improved safety compliance across project sites.

## Rationale

Traditional construction methods rely heavily on manual labor, which can lead to delays, variability in quality, and safety risks. Robotics and automation can bridge this gap by standardizing critical tasks, reducing defects, and minimizing human exposure to hazardous environments.

This challenge seeks to encourage innovation in robotics design, automation workflows, and safety integration, resulting in solutions that are practical, scalable, and transformative for the construction sector.

## Core Objectives

- 1. Enable faster project delivery through robotic automation.
- 2. Reduce defects with precision-driven robotic systems.
- 3. Improve safety compliance by minimizing human exposure to hazardous tasks.
- 4. Promote modular and scalable system design suitable for diverse construction projects.
- 5. Low maintenance suitable for Indian conditions (Humidity, Fluctuating power, High temp.)

# II. Targeted Participants and Eligibility

The challenge is open to participants across the construction and robotics ecosystem, including:

| Category                         | Description                                                                        |
|----------------------------------|------------------------------------------------------------------------------------|
| Startups (Early to Growth Stage) | Robotics, automation, and construction-tech startups developing scalable solutions |
| Independent Innovators           | Individual developers or innovators with functional prototypes or solutions        |
| Academic & Research Institutions | Universities, research labs, and incubators working on robotics and automation     |
| Technology Companies             | Firms with existing robotics or automation solutions ready for customization       |

### III. Scope of the Robotics Solution

Participants are expected to design solutions covering any of the following functional scope:

#### Core Functional Requirements (Indicative areas: Non-Exhaustive)

- Robotic Rebar Tying – Automated tying systems ensuring speed and consistency.
- Robotic Welding – Precision welding with defect detection and reduction mechanisms.
- Automated Concrete Pouring – Controlled pouring systems ensuring uniformity and reduced wastage.
- Safety Integration – Sensors, monitoring systems, and compliance mechanisms to enhance worker safety.
- Scalability & Modularity – Solutions adaptable to small, medium, and large-scale construction projects

Participants are free to suggest focus areas beyond the above but aligned to the core objectives.

### IV. Evaluation and Selection Criteria

Proposals will be evaluated by an expert committee based on the following criteria:

| Evaluation Parameter                          | Weightage  |
|-----------------------------------------------|------------|
| Innovation & Originality                      | 20         |
| Technical Completeness & Scalability          | 10         |
| User Experience                               | 20         |
| Real-World Applicability & Cost Effectiveness | 40         |
| Documentation & Demo Quality                  | 10         |
| <b>Total</b>                                  | <b>100</b> |

### V. Rewards & Adoption Pathway

## VI. Expected Deliverables

Participants must submit:

- A working robotics prototype or automation system.
- Source code repository with setup documentation (if software-driven).
- System architecture diagram.
- Demo video or live presentation.

## VII. Goal & Expected Impact

Aims to create a robust, inclusive, and sustainable robotics ecosystem that:

- Accelerates construction timelines.
- Reduce labour cost
- Improves structural quality and reduces defects.
- Enhances safety compliance across project sites.